

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester II)
Scala for Data Science Practical

Practical - 7

Roll No. : S006	Name:Seja Manohar Deshmukh
Class :SYIT	Batch:01
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Practical 07

Write a program to perform element-wise addition, subtraction multiplication and division of two Breeze matrices.

INPUT:

```
import breeze.linalg._

object S009 extends App{
  val A = DenseMatrix((1.0, 2.0, 3.0),(4.0, 5.0, 6.0))
  val B = DenseMatrix((6.0, 5.0, 4.0),(3.0, 2.0, 1.0))

  val add = A + B
  val sub = A - B
  val mul = A *: B
  val div = DenseMatrix.tabulate(A.rows, A.cols){ (i, j) => A(i, j) / B(i, j) }

  println("Matrix A:")
  println(A)
  println("\nMatrix B:")
  println(B)

  println("\nAddition :")
  println(add)
  println("\nSubtraction :")
  println(sub)
  println("\nMultiplication :")
  println(mul)
  println("\nDivision :")
  println(div)
  println("S006 Sejal Deshmukh")
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
```

Matrix A:

1.0 2.0 3.0

4.0 5.0 6.0

Matrix B:

6.0 5.0 4.0

3.0 2.0 1.0

Addition :

7.0 7.0 7.0

7.0 7.0 7.0

Subtraction :

-5.0 -3.0 -1.0

1.0 3.0 5.0

Multiplication :

6.0 10.0 12.0

12.0 10.0 6.0

Division :

0.16666666666666666 0.4 0.75

1.3333333333333333 2.5 6.0

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Process finished with exit code 0